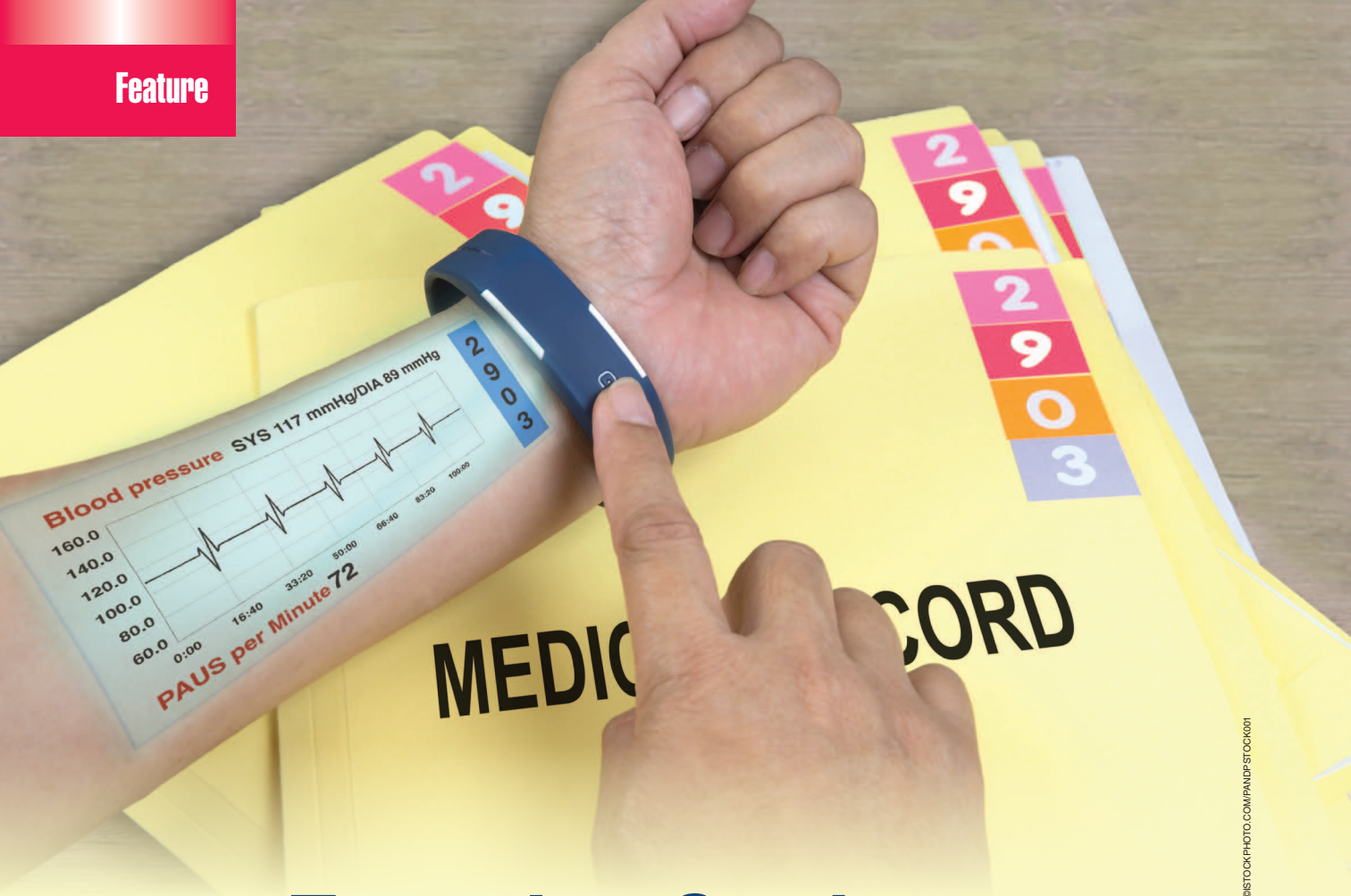




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Towards a Continuous Non-Invasive Cuffless Blood Pressure Monitoring System Using PPG: Systems and Circuits Review

Guoxing Wang, Mohamed Atef, and Yong Lian

Abstract

This paper presents an overview of the cuffless, continuous time, non-invasive blood pressure measuring devices (cNIBP) based on photoplethysmography (PPG). The cNIBP fundamental challenges,

such as motion artifacts (MAs), noise, low PPG signal amplitude with large DC components, and possible solutions are discussed. Techniques and circuits for MAs reduction, DC photocurrent cancellation, and sensitivity improvements are presented. To enhance the BP system performance, both the optimization of the integrated PPG receiver design and the reduction of the LED driver power consumption are introduced. Useful design considerations are proposed for designers to improve the PPG-based BP system accuracy.

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I. Introduction

Hypertension (or high blood pressure) is a major risk factor for dangerous health conditions such as heart attack, stroke, chronic heart failure, and kidney disease [1]. High Blood Pressure (BP) generally develops over many years without any symptoms, and could cause damage to blood vessels and heart continuously. According to WHO, worldwide high blood pressure causes 7.5 million deaths, about 12.8% of the total of all deaths [2]. Among people with high blood pressure, only 54% of them have their condition under control [3]. Thus, high blood pressure is one of the most important challenges in public healthcare globally.

To diagnose and manage high blood pressure, it is important to measure BP routinely. Although the traditional clinical BP measurement provides information on one's BP level, it comes with several shortfalls, e.g. inconvenience, sparse BP readings, and white coat syndrome. Studies show that risks associated with hypertension can be better managed through frequent BP monitoring. BP monitoring at home not only provides better prognostic value but also prevents from serious organ damage compared with clinic BP monitoring [4]. This is because in-home BP monitoring gives more frequent BP readings than clinic BP monitoring. The frequent measurement leads to more accurate BP data than sparse clinic BP measurement. Additional benefit of home BP monitoring is that it helps to identify white coat and masked hypertension. Compared with ambulatory BP monitoring, in-home BP monitoring costs less and is easy to implement.

There are two different methods for BP monitoring, one is cuff-based and the other is cuffless. Commercially available BP devices based on the oscillometric method use an inflatable cuff during measurement. However, the inflatable cuff is not feasible for long-term BP measurements due to discomfort caused by repeated inflation and deflation. If used during sleep, it can cause sleep disturbances [5], [6].

Volume clamp-based BP devices can measure continuous BP using an inflatable finger cuff with a built-in PPG (photoplethysmography) sensor and a pressure controller unit [7]. When the volume under the finger cuff becomes constant (constant PPG signal), the BP value can be evaluated as the pressure value in the finger cuff. However, the vessel under the finger cuff is pressed, and the oxygen level decreases in these vessels, which can cause tissue hypoxia during continuous measurements. Variations in finger positioning within the cuff influence the BP measurement, reducing its precision [8].

The tonometry method is suitable for measuring instantaneous BP. It applies a small force perpendicular to a superficial artery [9], [10]. A pressure transducer measures the pressure value at the contact point. The external force must be small to not dramatically reduce the blood flow through the artery. The contact is maintained, and the transducer measures the blood pressure wave. The tonometry method needs a continuous control for the contact pressure and the transducer, which increases the system complexity and cost. The accuracy is affected by the motion artifact. The BP monitoring becomes uncomfortable as the measurement time increases.

Pulse wave velocity (PWV) is a cuffless method that can measure beat-to-beat BP over a long period [11]. PWV refers to the velocity of the pressure pulses generated during left ventricular ejection, after opening of the aortic valve. The device normally measures pulse transit time as the time delay between the carotid artery pulse wave and the femoral artery pulse wave and use that to calculate the PWV. Unfortunately, it requires a trained medical expert to locate the arteries used for the BP measurements. The accuracy is significantly affected by the motion during BP measurement. Also, the distance between locations to determine the arrival time of the pulse wave must be measured precisely and manually [12].

Ideally the best method for continuous non-invasive blood pressure (cNIBP) monitoring should offer high accuracy, remain comfortable for long time measurements (i.e. cuffless measurement), require no medical supervision, and automatically store the data for later usage. The recent development in wearable wireless cuff-less continuous BP monitoring devices meet most of these requirements and could become the best candidate of cNIBP.

Due to the emerging interest in recent years in developing cNIBP measuring devices, IEEE introduced a cuffless BP measurement standard, i.e. IEEE standard 1708–2014, which standardizes “Wearable, Cuffless Blood Pressure Measuring Devices” [5]. Table I summarizes the recommended accuracy levels that are equivalent to the American Association for the Advancement of Medical Instrumentation (AAMI) requirements [13] and the BHS protocols that are recommended for cuff-based devices [5]. The manufacturer should determine the grade level of the Device Under Test (DUT) according to the accuracy levels indicated in Table I. The BP mean absolute difference (MAD) is calculated using the DUT and a reference device for both systolic blood pressure (SBP) and diastolic

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blood pressure (DBP). The British Hypertension Society (BHS) requires the accuracy for BP MAD to be less than ± 5 mmHg, with a standard deviation less than 8 mmHg.

Most of wearable BP devices under development are based on PPG or PTT (Pulse Transit Time) techniques, thus require no cuff and no medical supervision. They are low cost and comfortable to use. These devices could pave the way for wide adoption of home BP monitoring. The PPG/PTT BP-based methods still have some challenges with respect to their accuracy and the need for recalibration. They cannot reach Grade A level for BP standard requirements. More hardware and software improvements are still needed for a wearable cNIBP device that can function in daily life. The main challenges and issues like motion artifacts (MA), large background light, in-band noise, and low-power operation affects the BP measurement accuracy and portability.

This paper presents the current challenges and the possible solutions in the development of cNIBP devices. The available sub-blocks and the algorithms developed for cNIBP devices are discussed. Recommendations and guidelines are provided for the improvement of each sub-block toward a high-performance cNIBP device.

The rest of the paper is organized as follows. The PPG sensing principles of operation are introduced in Section II. The different BP models and algorithms for calculating BP values are presented in Section III. The PPG sensor circuitries are discussed in Section IV. Commercially available cNIBP devices are provided in Section V and conclusion remarks are drawn in Section VI.

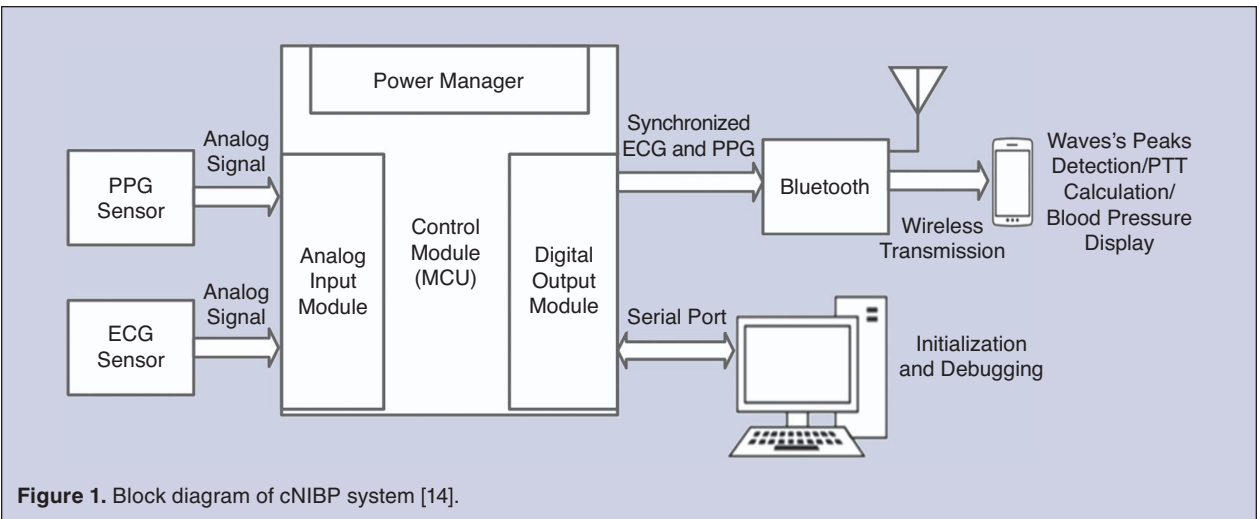
II. Photoplethysmography (PPG) Signal Acquisition

Two types of cuffless BP measurement technique are widely used in wearable BP devices, i.e. PPT based or PPG-only techniques. For PTT based devices, both ECG and PPG signals are recorded simultaneously. Rich features can be extracted from the combination of ECG and PPG signals, which increases BP model accuracy. More details about BP modeling using PTT will be covered in Section III A.

Figure 1 shows the block diagram of a typical cNIBP system. This hardware unit acquires the ECG and PPG signals of the patient via a PPG sensor and an ECG sensor [14]. The MCU processes the input signals to calculate the BP. During the calibration phase, the MCU acquires an additional signal from a reference commercial blood pressure monitoring device. There is a communication channel between the

Table I.
MAD accuracy level with comparisons to the ANSI/AAMI SP10 and BHS evaluation systems [5].

MAD(mmHg)	ANSI/AAMI SP10	BHS	Recommended Grading
≤ 4	pass	Grade A	A
4–5	pass	mostly in Grade A, less in Grade B	A
5–6	pass or fail	mostly in Grade B, less in Grade A, extremely less in Grade C and Grade D	B
6–7	mostly fail, less pass	mostly in Grade C, less in Grade B and Grade D	C
≥ 7	fail	worse than Grade C	D



cNIBP and PC/smartphone via serial port/wireless so that the ECG and PPG signals and BP values from the user can be collected by the phone/PC for monitoring and storage. The smartphone can transmit the collected data and report it to a medical care unit.

A PPG sensor is required to sense the light signal modulated by the BP wave traveling through the arteries. The PPG sensor consists of two components, a light emitting diode (LED) as a light source and an optical receiver. There are two modes for PPG sensing, transmission mode and reflectance mode as shown in Fig. 2. Light passing through human tissue can be absorbed by skin, bone, and arterial and venous blood, as shown in Fig. 3. The optical receiver detects the changes in the reflected light intensity due to the change in the blood flow volume. The incident light is amplitude-modulated by the changes in blood volume during the cardiac cycle, with maximum blood volume occurring in the systolic phase and a minimum blood volume at the diastolic phase. The systolic peak (primary peak) is coming from the direct pressure wave traveling from the left ventricle to the periphery of the body, the diastolic peak (secondary peak) is the effect of the pressure wave reflections by arteries of the lower body [15], as shown in Fig. 4. The received optical signal has both AC and DC components. The AC component has an amplitude that depends on the amount of light absorbed by the change in the blood volume and has a fundamental frequency related to the heart rate. The other detected optical signal component has a constant value (DC) depending on the tissue structure thickness, skin color, and the average blood volume. This fixed tissue structure absorbs a constant amount of light and does not change with time [16].

A. The Selection of PPG Sensing Mode

As mentioned earlier, there are two modes for PPG measurement. The transmission mode requires a higher LED light intensity and works well for thin body parts, whereas reflectance mode shows a higher detected signal level compared to transmission mode at low transmitted light intensity in thicker body parts. In general, reflectance mode can give the same signal quality as transmission mode using lower LED power [18]. As the probability is much higher for weak light photons to be reflected than transmitted through thick tissue, reflectance mode can be used in any part of the human body, while transmission mode is limited to peripheral parts such as the fingertip, foot, and earlobe. For example, PPG signals from the front of the head and chest can only be detected in reflectance mode. It is worth noting that fingertips and earlobes have limited blood perfusion such that these sites are susceptible to low ambient temperatures; addi-

tionally, fingertip sensors are affected by daily activities. Thus, the reflectance signal from the wrist or forehead would be a better solution for wearable devices. However, the signal received from reflectance mode is easily affected by MAs and pressure disturbances, which limit the measurement accuracy [17], [18].

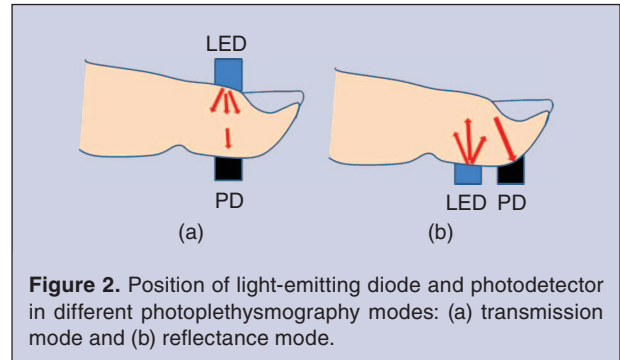


Figure 2. Position of light-emitting diode and photodetector in different photoplethysmography modes: (a) transmission mode and (b) reflectance mode.

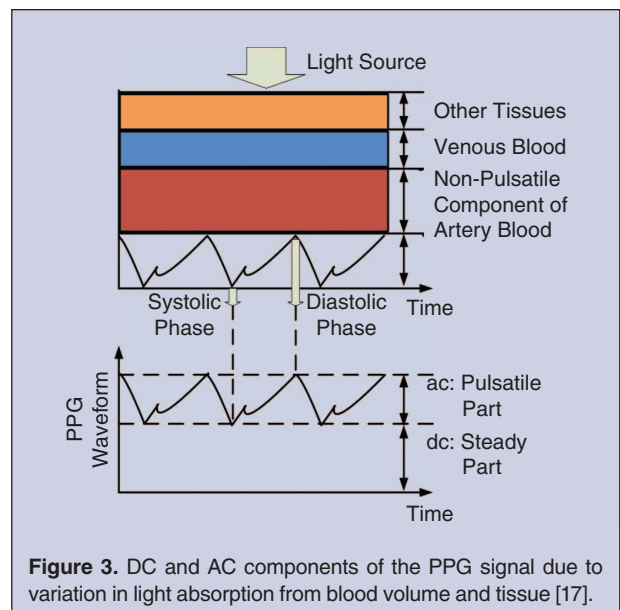


Figure 3. DC and AC components of the PPG signal due to variation in light absorption from blood volume and tissue [17].

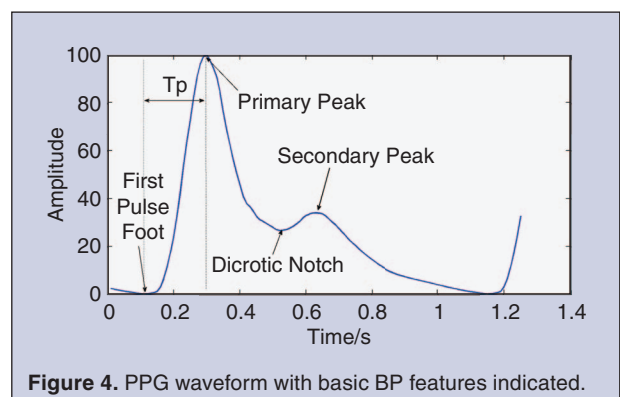


Figure 4. PPG waveform with basic BP features indicated.

To minimize the loss in thick tissues, the tissue thickness at the PPG sensor position should not be greater than the penetration depth of the light wavelength.

B. Light Wavelength

The light penetration depth is inversely related to its absorption coefficient (α). The greater the absorption, the smaller the light penetration depth in tissue. To minimize the light loss in thick tissues, the tissue thickness at the PPG sensor position should not be greater than the penetration depth of the used light wavelength. For visible light, the penetration depth through tissue under the skin surface ranges from 100 μm to 1200 μm [19]. A reflectance PPG sensor located in thin tissue regions like fingertips, earlobe, and the forehead results in stronger signals than in thicker tissue regions such as the chest. Red and IR light can penetrate deeper into the tissue than green light wavelengths can; these wavelengths can thus give better results for deep-tissue blood flow in the muscles. In addition, the photodetector responsivity of the red and IR wavelengths is greater than that of the green light. Green light, on the other hand, is absorbed near the surface, so it is suitable for the shallow blood flow directly under the skin. Short wavelengths like blue and green (405 to 520 nm) have stronger blood absorption than red and IR light (650 to 850 nm). Therefore, the signal-to-noise ratio (SNR) for a green light source is better than that of an IR source when used for superficial blood flow in the skin. The received signal at a certain light penetration depth can be controlled by adjusting the distance between the LED and the photodiode. By increasing the distance between the photodiode (PD) and the LED, the PD will detect the light rays with deeper penetration. The closer the

PD and the LED are, the better shallow surface light can be detected. The practical distance between the PD and LED is from 4 to 6 mm [18]. Green light has better immunity to MAs than red and IR lights, as the IR light penetrates more deeply and is affected by all surface and deep tissue movements [20]. The PPG signal from the green light is reduced sharply by increasing contact pressure compared with the IR PPG signal. This is an indication of the occlusions of superficial skin blood vessels. The PPG secondary peak and dicrotic notch are more structured with red than with green light (Fig. 4) [19].

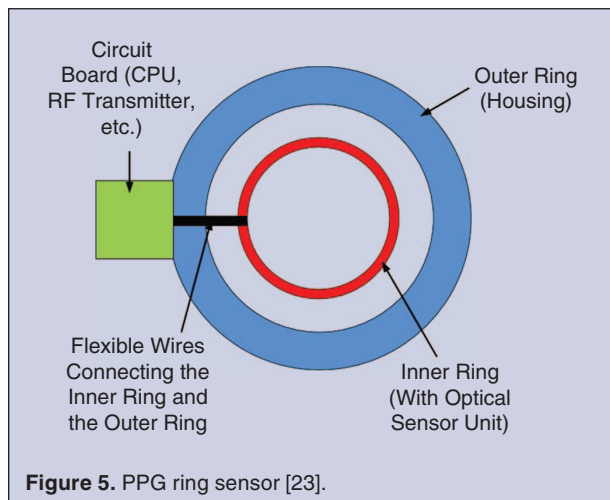
C. PPG Sensor Position

There are different possible sites for PPG sensor placement, such as the fingertip, forehead, earlobe, chin, and wrist. Fingertips give the strongest PPG signal, the forehead and earlobe have good signal strength, and the wrist has a low signal amplitude. Although the fingertip produces the strongest PPG signal, the MAs associated with routine activities produce a great deal of noise, which reduces the signal quality. Wrist and earlobe locations also suffer from MAs due to hand and neck movements, respectively. The forehead has a flat and large region for sensor placement, easy attachment by using a head bandage, and is a stable region with low motion artifact effects. It also produces a strong PPG signal. However, as forehead PPG sensors work only in reflectance mode, dirt and sweat may affect measurement accuracy. The chin region has a fair PPG signal, but it suffers from growing hairs, dirt and sweat [18], [21], [22]. The work in [18] studied the different locations with respect to PPG signal strength, attachment, artifact, and portability. They introduced a decision matrix to determine the preferred sensor locations, and the forehead received the highest score.

D. PPG Sensor Structures

A PPG sensor position and structure should be comfortable enough for use during daily activities. Fingers can be easily accessed, and they provide a strong PPG signal. One of the finger-based structures is a ring sensor, which can be attached to the finger base, as shown in Fig. 5 [23]. To reduce the MAs, a double ring structure was developed to reduce the effect of outside forces.

A wristwatch-type PPG sensor was introduced in [24]. This design collects the PPG signal from both the



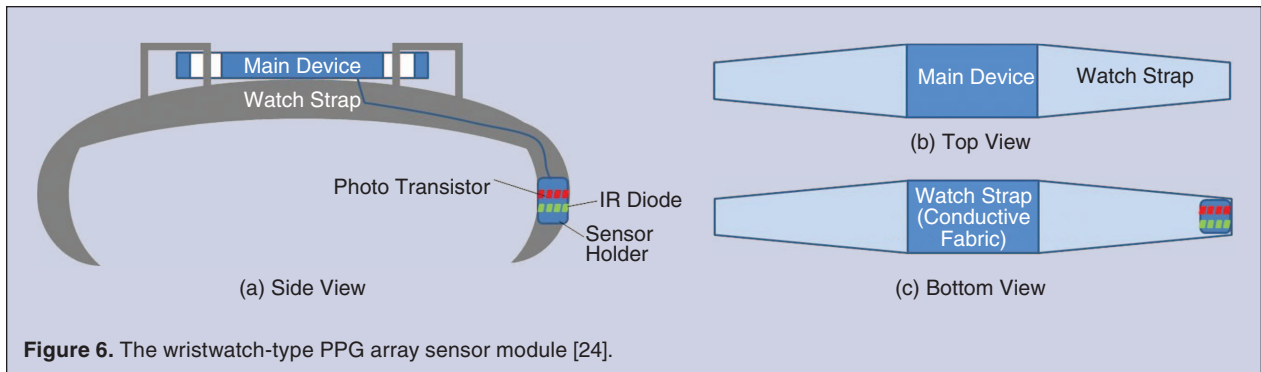


Figure 6. The wristwatch-type PPG array sensor module [24].

wrist radial and ulnar arteries. The wristwatch is comfortable to wear, but the **PPG signal coming from the wrist area is weak**. An array of phototransistors and LEDs were placed to improve the PPG sensor sensitivity and its accuracy. A conductive fiber wristband was employed to reduce external noise. An armband-based wearable monitoring device was developed in [25] using a near-infrared LED and a phototransistor in reflective mode.

A Nellcor forehead reflectance-mode sensor [26] secures the sensor by means of a headband. Integrating the PPG sensor into preexisting equipment like a soldier's helmet provides a comfortable and acceptable attachment for the user [18].

In [22], an ear-worn reflective PPG sensor with improved surface coupling was proposed. The sensor has been proven to be robust against pressure-induced sensitivity degradation [22]. A wireless earpiece PPG sensor with a magnetic earring was introduced in [27]. The sensor can be comfortably worn on the earlobe, and it contains an embedded accelerometer to provide MA noise reference for adaptive noise cancellation. However, the earlobe becomes susceptible to inaccuracies due to peripheral vasoconstriction and perfusion. To overcome these limitations, the external auditory canal has been proposed as an alternative monitoring site, on the hypothesis that this central site will be better perfused [28].

For low cost and high integration ability, blood pressure can be estimated using a smartphone camera. Initial BP screening with a low-cost solution using a simple and portable device is the aim for such smartphone health-care applications [29]–[31]. The PPG signal is processed by an artificial neural network (ANN) built in a smartphone to evaluate the BP, as shown in Fig.7 [31]. The on-line evaluation of the experimental BP results of the proposed method shows a maximum error of 12 mmHg [31]. **This reported BP determination accuracy using a smartphone is still worse than the 8 mmHg required for it to be used as a commercial healthcare device.**

III. Blood Pressure Modeling and Algorithms

There are several methods to estimate the BP depending on the sensing and calculation techniques. One method is to calculate BP based on the PTT. The Another method uses PPG signal only to calculate the BP in order to reduce the hardware complexity.

There are two main directions for BP estimation using PPG only. The first one involves the parametric models, which try to extract specific parameters such as instantaneous heart rate, systolic, and diastolic periods from individual PPGs. Then, the BP can be predicted using these extracted parameters and a parametric model (Section III B). The second direction is the non-parametric models. Non-parametric BP methods try to extract specific features (time domain or frequency domain features) with strong relations to BP from the PPG waveform.

The main steps to calculate the BP from PPG signals are listed in [32]; some steps can be merged from different algorithms into the following steps:

- 1) Preprocessing: deals with noises caused by power line interface, baseline wandering [33], motion artifacts [34]–[37], and signal quality assessment [38], [39]. Classification is performed to discriminate between motion artifacts and clean PPG. A band pass filter is used to remove the high-frequency noise above 5 Hz and reject the DC and low-frequency noise component below 0.5 Hz.

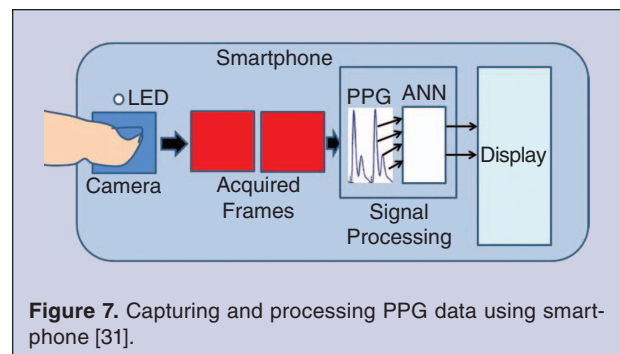


Figure 7. Capturing and processing PPG data using smartphone [31].

- 2) Feature Selection: selects a set of features that are the most relevant and least redundant for BP, to enhance the performance of the BP classification model and reduce the algorithm complexity by selecting a smaller number of efficient features [40].
- 3) Feature Extraction: extracts signal features from time and frequency domains [30], [41], [42]. Time domain characteristics can be defined as (a) pulse amplitude; (b) pulse length (ON time); (c) pulse rate (peak-to-peak time); (d) valley depth difference (voltage difference between two minimums); and (e) peak height difference (between two maximums) [32], as shown in Fig. 10 and Fig 8. Frequency domain characteristics of the PPG signal can be extracted by applying the Discrete-time Short Time Fourier Transform (STFT).
- 4) Regression: estimates the blood pressure with a proper model, such as parametric PTT-based regres-

sion [14], machine learning [43], [44], the Windkessel model [45], or the BP transport model [46], etc.

A. PTT-based Blood Pressure Models

Pulse Transit Time (PTT) is the time it takes the Pulse Pressure waveform to propagate through a certain length of the arterial tree. The PAT is the time delay that takes for the pulse wave to travel along the artery and arrive at the periphery in the same cardiac cycle. The PAT can be measured as the time delay between the R-wave peak of the ECG signal and a particular point of the PPG signal (foot PAT_f, peak PAT_p or maximum slope point PAT_s) as shown in Fig. 9 [47]. The PAT is equal to the summation of the PTT and the pre-ejection period (PEP). The PEP is the time to open the aortic valve to start pushing blood out of the ventricle. As the PEP is difficult to measure from the PPG signal, the PAT is acceptably approximated to be measured as the PTT. Both PPG and ECG waveforms are required for a PTT-based BP device.

PWV has been found to be an important parameter in the analysis of arterial blood pressure. The propagation of blood in the artery is very similar to that of the propagation of a compressible fluid. The relationship between vessel elasticity and the wave speed of a pressure pulse in a thin-walled vessel can be described by the Moens-Kortweg equation ([48]):

$$PWV = \frac{L}{PTT} = \sqrt{\frac{hE_0 \exp(\zeta P)}{\rho R}}, \quad (1)$$

where PWV is the pulse wave velocity, L is the length of the vessel, PTT is the pulse transit time, h is the thickness of the vessel wall, R is the radius of the vessel, ρ is the blood density, E_0 is the zero-pressure modulus in mmHg, and ζ is a constant that depends on the particular vessel (typically 0.016 mmHg^{-1} to 0.018 mmHg^{-1}).

Under the assumption that the overall percentage change in the arterial wall thickness (h) and arterial radius (R) are small over the physiological pressure range, equation (1) can be simplified to relate BP to PTT [49], i.e.:

$$P = \frac{1}{\xi} \left(\ln \left(\frac{2R\rho L^2}{hE_0} \right) - 2 \ln(PTT) \right) = k_1 \ln(PTT) + k_2, \quad (2)$$

where k_1 and k_2 are patient-dependent coefficients.

According to the works of [50], within a small pressure range, the $\ln()$ function can be linearized, and the BP equation becomes the following:

$$P = \frac{1}{\xi} \left(\left(\frac{2R\rho L^2}{hE_0} \right) \left(\frac{1}{PTT^2} \right) - 1 \right) = \frac{A}{PTT^2 + B}, \quad (3)$$

where A and B are patient-dependent constants.

By using series expansion, the $\ln(PTT)$ function in equation (2) can be approximated by the linear model in equation (4), which has also had been proved empirically

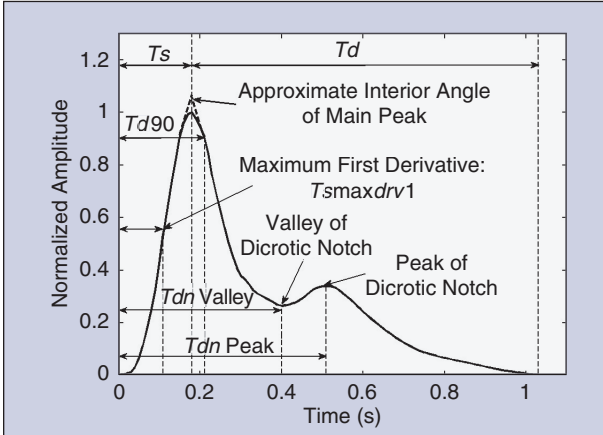


Figure 8. PPG waveform with important BP features indicated.

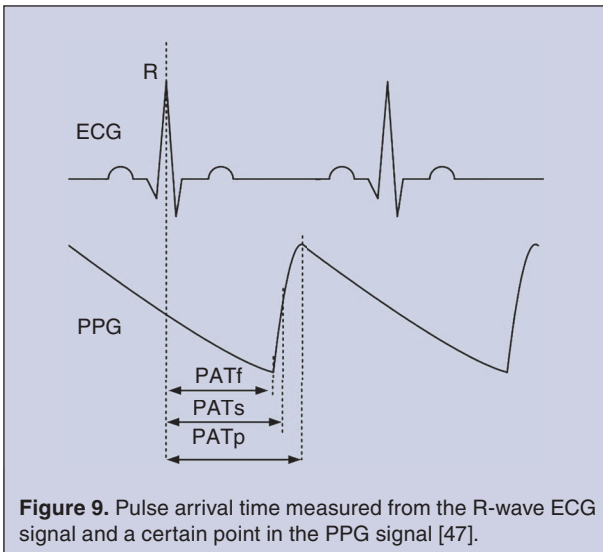


Figure 9. Pulse arrival time measured from the R-wave ECG signal and a certain point in the PPG signal [47].

[51]. Because of its simplicity and low computational complexity, it has become a widely used model [52], [53].

$$P = \alpha \cdot \text{PTT} + \beta, \quad (4)$$

It was observed in [47] that BP is highly correlated not only with PTT but also with heart rate (HR); hence, they proposed the following model:

$$P = a_1 \text{PTT} + b_1 \text{HR} + c_1, \quad (5)$$

A non-linear approach is proposed by [54], [55]. It is based on the non-linear relationship between the pressure and the vascular volume.

Shown in (6) at the bottom of the page, where L is the distance traveled by the pulse, b is a patient-dependent coefficient, and $P_{\text{hydro}} = \rho gh$.

There are numerous models proposed by the researchers. However, the accuracy of these models very much depends on how accurately the unknown model parameters can be estimated. Most of these unknown parameters are patient dependent. Hence, the model should be tailored for each patient in order to monitor his or her blood pressure continuously and accurately. In addition, for a given patient, these model parameters may change with time, and thus, subsequent calibration may be needed to maintain accuracy over time [56].

B. PPG-only Blood Pressure Models

The PPG-only models reduce the hardware complexity for cuff-less BP estimation. There are two main directions in using PPG-only approach. The first one involves the parametric models, which try to extract specific parameters such as instantaneous heart rate, systolic, and diastolic periods from individual PPGs. Then, the BP can be predicted using these extracted parameters and a parametric model. Some examples of the parametric methods are the Windkessel model [45], regression of long-term and short-term features [57], and the pulse transport theory-based model [46].

Instead of directly relating systolic and diastolic BP values with PPG features, the Windkessel model first maps PPG features to specific body parameters and then derives BP values from these parameters [45]. The 2-element Windkessel model estimates the total peripheral resistance (R) and arterial capacitance(C) of the body using PPG features. Then, a BP model using lin-

ear regression can simulate the arterial blood pressure equations (7) and (8) [45].

$$P_s = P_{t|t=T_s} = P_{ts} e^{-\frac{T_s}{RC}} + \frac{I_0 T_s C \pi R^2}{T_s^2 + C^2 \pi^2 R^2} (1 + e^{-\frac{T_s}{RC}}), \quad (7)$$

$$P_d = P(t|t=T_d) = P_{td} e^{-\frac{T_d}{RC}}, \quad (8)$$

where P_{ts} and P_{td} are the initial values of systolic P_s and diastolic P_d blood pressure, respectively, and I_0 is the maximum value of the blood pumped by the heart during systole. Similarly, R and C can also be expressed as a function of P_s, P_d, T_s, T_d and I_0 .

The parametric models can achieve good prediction results an individual [45], [46], [57], but the accuracy decreases over time. Moreover, these over-simplified parametric models require initial calibration and frequent recalibration for each person.

The second direction is the non-parametric models which try to extract specific features, with strong relations to BP, from the PPG waveform, refer to Fig. 8 and Table II. Then, the BP can be predicted using these

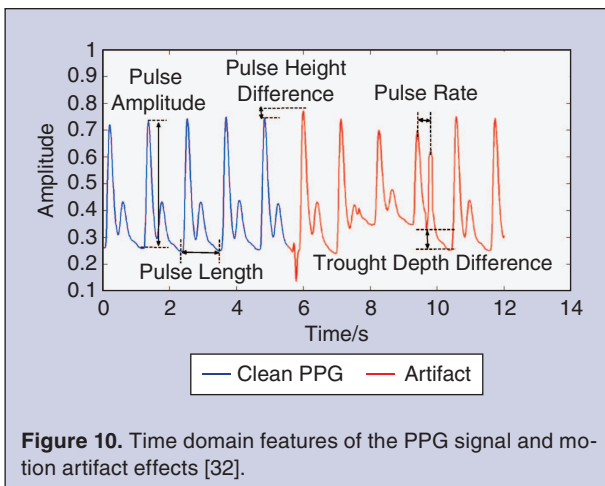
Table II.
Important PPG BP features indicated in Fig. 8.

Features	Definition
f1	T_s systolic time
f2	T_d diastolic time
f3	T_{dnPeak} time distance from onset to Dicrotic Notch
f4	T_{d90} the time distance from onset to the point with 90% of the maximum height within diastolic time(DN)
f5	$T_{smaxdn1}$ time distance from onset to the point with maximum first derivative within T_s
f6	amplitude at valley of DN
f7	amplitude at peak of DN
f8	approximate interior angle of main peak
f9	mean value of whole pulse
f10	mean value of pulse within T_s
f11	mean value of pulse before valley of DN
f12	mean value of pulse after valley of DN

$$P = \begin{cases} \frac{L\sqrt{\rho b}}{\sqrt{1 + \exp(bP_{\text{int}})}}, h = 0 \\ \frac{2L}{gh\sqrt{\rho b}} \cdot \ln \left| \frac{\sqrt{\exp[b(P_{\text{int}} - P_{\text{hydro}})] + \exp(-bP_{\text{hydro}})} - \sqrt{\exp(-bP_{\text{hydro}})}}{\sqrt{\exp[b(P_{\text{int}} - P_{\text{hydro}})] + 1} - 1} \right|, h \neq 0 \end{cases} \quad (6)$$

Table III.
Comparison of different PPG-based BP models and algorithms.

	Algorithms	Dataset	Calibration	Accuracy(mmHg)
[58]	Non-linear Regression, (ECG + PPG)	5 Case	A certain time for each person	ME: SBP(5.4%), DBP(7.7%)
[59]	Linear Regression, (ECG + PPG), adaptive Kalman filter	10 subjects	Every 3.3 hours	STD: SBP(1.69), DBP(1.04)
[37]	ACF, linear regression, (ECG + PPG)	MIMIC, 100 Cases	A certain time for each person	ME $\pm$ STD: ABP(3.06 $\pm$ 3.69)
[34]	Iterative linear, (ECG + PPG)	MIMIC, 6 Cases	A certain time for each person	ME $\pm$ STD: SBP(0.13 $\pm$ 3.73), DBP(0.22 $\pm$ 4.33 mmHg)
[25]	Moens-Korteweg, (ECG + PPG)	10 Cases	Once for every 12 hours	ME $\pm$ STD: SBP(2.4 $\pm$ 5.7)
[41]	SVR, (PPG)	MIMIC	Never	MAD $\pm$ STD: SBP(12.38 $\pm$ 16.17), MAP(7.52 $\pm$ 9.54 mmHg), DBP(6.34 $\pm$ 8.45 mmHg)
[42]	ANN (PPG)	MIMIC, 5000 Cases	Never	MAD $\pm$ STD: SBP(3.24 $\pm$ 4.75 mmHg), MAP(2.16 $\pm$ 3.14), DBP(1.79 $\pm$ 2.70)
[30]	SVM, (PPG)	UQ, 32 Cases	Never	Classification(20 interval): SBP(98,81 %), DBP(98.21 %)
[43]	Random Forest, (PPG)	410 Cases	Never	MAD ($\leq 5,10,15$): SBP(62.43%, 86.34%, 90.37%), DBP(79.02%, 90.24%, 93.90%)
[60]	Linear regression, (PPG)	18 subjects	A certain time for one person	MAD $\pm$ STD: SBP(sleeping: 2.68 $\pm$ 0.39 mmHg, sleepy: 4.37 $\pm$ 1.20)
[45]	Windkessel model, (PPG)	UQ, 32 Cases	Once for all	MAD $\pm$ STD: SBP(0.78 $\pm$ 13.1 mmHg), DBP(0.59 $\pm$ 10.2)
[46]	Harmonic Balancing model, (PPG)	5 Cases	A certain time for one person	MAD $\pm$ STD: SBP(1.37 $\pm$ 7.61 mmHg), DBP(-1.40 $\pm$ 6.00)



extracted features. Non-parametric models try to solve the low accuracy of parametric models.

Examples of non-parametric learning methods can be found in [30], [41]–[43]. Reference [30] tries to classify the SBP and DBP within different prediction ranges with 19 features extracted from the PPG signal using only a sup-

port vector machine (SVM). The estimation accuracy can reach 98.81% and 98.21% for SBP and DBP, respectively. Reference [41] gets its features from both the PPG and ECG signals. Using support vector regression (SVR), it reaches grade B for both DBP and mean blood pressure (MBP) estimations and reaches grade C for SBP according to the BHS protocol. In [42], an ANN-based learning process is used to train the features from the PPG signal only. The results can fulfill the AAMI requirement. Using only the PPG signal, [43] implements a random forest (RF)-based prediction model and achieves grade B for both SBP and DBP by BHS. Even though the non-parametric BP models exhibit better accuracy than the parametric models, they suffer from heavy calculations.

Table III compares the different PPG-based BP models and algorithms. The first five methods in Table III [25], [34], [37], [58], [59] use both ECG and PPG signals with linear or nonlinear regression. These methods have higher accuracy, but they require both ECG and PPG signals and frequent calibration. The methods in [30], [41]–[43] do not need frequent calibration, and only PPG information is needed. However, they require excessive training and

calculations for the SVM, SVR, and ANN algorithms and show higher error than the other methods using both ECG and PPG signals. The last three methods [45], [46], [60] are only based on the PPG signal. They use linear regression and non-parametric models that do not need as much training and calculation as the SVM methods. However, they do require calibration and have the highest error.

C. Motion Artifact (MA) Cancellation

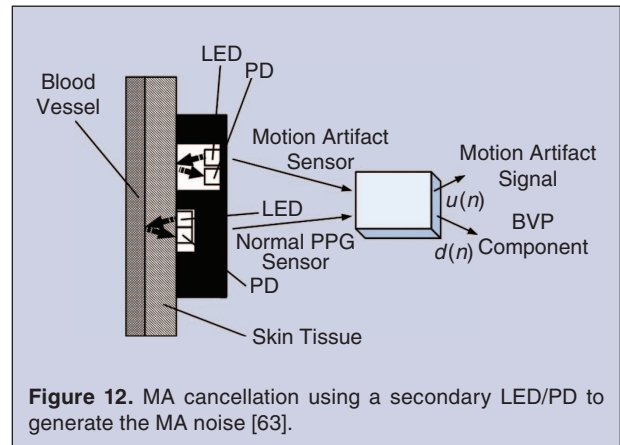
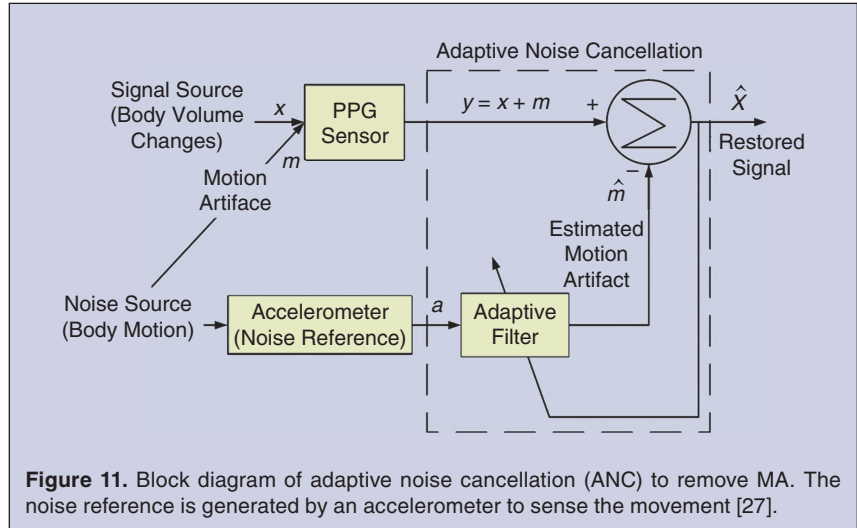
Cancellation

MAs introduce a disturbance to the PPG signal. One of the examples is as shown in Fig. 10. The time domain parameters like pulse amplitude, pulse length, pulse rate, valley depth difference, and peak height difference, will vary too much at strong MA events. A wearable BP device must be able to reduce/cancel artifacts generated by movements. The high-frequency MA components can be canceled by a 5 Hz LPF, but the in-band MA noise cannot be removed by a simple filter.

Advanced signal processing techniques are required to deal with MA disturbances. If the motion-induced noise signal is known, the desired noise-free signal can be obtained by subtracting the MA noise from the noisy output of the PPG sensor, as shown in Fig. 11. The adaptive filters require a MA reference noise signal that has a strong correlation to the MA and is uncorrelated with the MA-free PPG signal. A synthetic MA noise reference signal can be generated from the MA-corrupted PPG signal [61]. Reference MA noise signals can be obtained from an additional transducer attached to sense the movement (accelerometer) [27], [62] or by employing an additional reflectance-type optoelectronic sensor, as shown in Fig. 12 [63]. Fig. 12 shows the principle of MA cancellation using a secondary LED/PD to generate the MA noise. The main sensor LED/PD contacts the skin to detect the blood volume pulse (BVP) component (inevitably MA affected). The secondary LED/PD does not contact the skin (7.5 mm air gap) to detect only MA [63]. Then, the MA is canceled by a non-linear adaptive canceler filter using a recursive least squares (RLS) algorithm.

IV. PPG Sensing Circuitries

The PPG receiver receives the weak optical signal, converts it, and amplifies it to a strong electrical signal. The optical signal is converted to electrical current by means of a large-area photodiode (a few mm²) to collect most of the PPG optical signal. A low-noise, high-gain preamplifier follows the PD to amplify the weak photocurrent to a



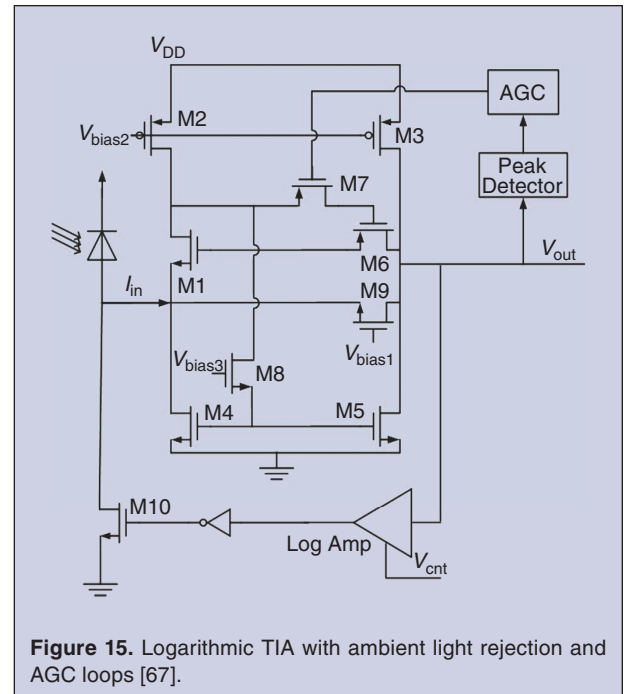
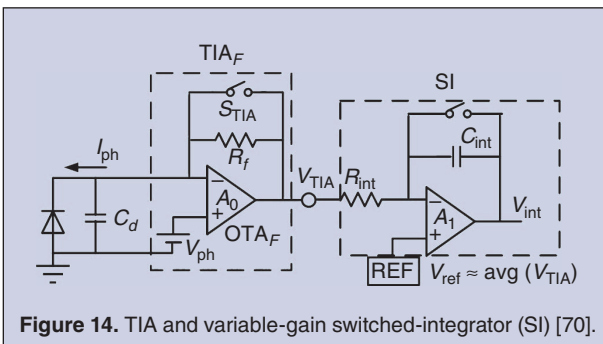
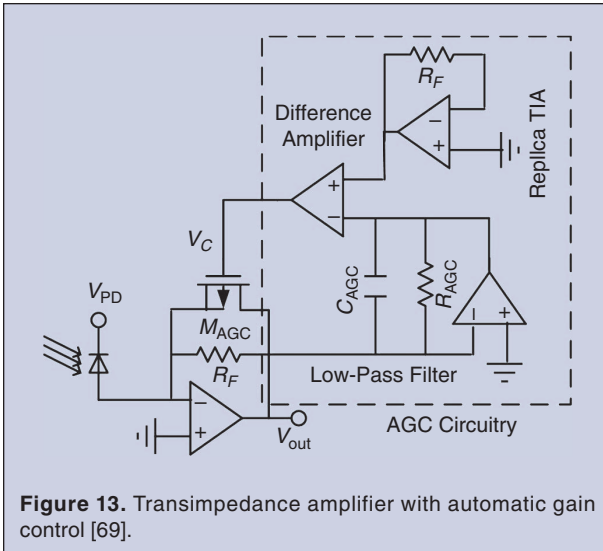
strong voltage. The main requirements of a PPG preamplifier are high sensitivity, high gain, wide dynamic range, and the ability to reject ambient light (DC photocurrent). A low power consumption is essential to prolong the battery life for portability. As the transmitted optical power is limited by safety regulations, the photodiode current can be as small as a few tens of nano-amperes. A high sensitivity (low noise) optical receiver is required to deal with this weak light signal. Depending on the relative positions of the receiver and the LED and the sensing location, the signal strength can vary dramatically from a few nA to several μ A. A wide dynamic range is a key parameter in order to accommodate the different signal levels. The light penetrates through the human tissues and is absorbed by the blood, and the light is then reflected after it has been modulated by the blood volume change. The modulation depth is very small, and the AC to DC ratio can vary from 0.1% to 3% depending on skin color, thickness, sensing position and the light wavelength. A DC photocurrent rejection is needed to prevent PPG receiver saturation.

A. Transimpedance Amplifier (TIA)

The TIA is the most important block in the PPG receiver as it determines the receiver sensitivity and bandwidth. The major limitation of the TIA is the high **parasitic capacitance** associated with the large-area photodiode, which deteriorates the gain, bandwidth, power consumption, and noise performance of the preamplifier. Shunt-feedback TIAs have succeeded in achieving high bandwidth and good sensitivity. However, they suffer from trade-offs between bandwidth, transimpedance gain, power, and sensitivity. Open-loop TIAs input stages such as common gate (CG) [64] and regulated cascode (RGC) show low input impedance [65], which can reduce the effect of the high input capacitance. However, the noise performance of these topologies is worse than that of feedback TIAs. Several techniques have been demonstrated in [66]–[68] to relax the effect of the large photodiode capacitance on the TIA performance.

The dynamic range of a TIA is defined by its overload current and its sensitivity. A TIA with a fixed transimpedance gain is not sufficient for the wide dynamic range required. There are two techniques to extend the TIA dynamic range, including automatic gain control (AGC) and

a second technique using signal level compression. An AGC is needed to improve the TIA linearity; for large input signals, the gain is reduced to keep the TIA working in the linear region. For a shunt-feedback TIA, the dynamic range can be extended by using a variable resistor changing with the input optical signal power, shown in Fig. 13. The variable feedback resistor can be implemented with a MOSFET operating in the linear region, connected in parallel with a fixed resistor to improve the linearity and to limit the maximum resistance (Figure 13). The AGC output V_c is a DC signal with a value depending on the input signal strength. The AGC can be implemented with a circuit that determines the output signal strength (peak detector or average power) to control the gate voltage of the MOSFET (a low-pass filter can perform the averaging process). The replica TIA generates a reference voltage for the difference amplifier. The difference amplifier subtracts the reference voltage from the averaged signal to give the control signal V_c [69]. The stability of the AGC-TIA is an important issue. When the feedback resistance (R_F) varies while keeping the open-loop gain fixed, both the bandwidth and the quality factor will change. If R_F is reduced, the frequency of the first open-loop pole at the input $1/(R_b \cdot C_T)$ increases, and it will come closer to the second pole at the output $1/(R_{out} \cdot C_{out})$, which may lead to peaking at a given fixed loop gain. The bandwidth of the AGC-TIA tends to increase with the magnitude of the input signal. The low cutoff frequency is determined by the integration time of the integrator. The low cutoff frequency should be as low as 0.5 Hz not to lose any of the PPG signal features. So, the capacitor C_{AGC} and the



resistor R_{AGC} should have a high value, which could be limited by the chip area.

The receiver in Fig. 14 is an example of using AGC. The first block is a conventional TIA with 500 K Ω resistive feedback [70] followed by a voltage mode switched-integrator (SI). This block serves multiple purposes, acting as a variable-gain amplification stage that enables implementation of large on-chip gains exceeding 50 M Ω , which can be easily tuned via the digitally controlled integration time.

A logarithmic TIA depicted in Fig. 15 uses compression techniques to reduce the signal level at high input current [67]. Additionally, an AGC loop is introduced to further extend the dynamic range. M1 acts as the logarithmic amplifier transistor. M6, M7 and M9 provide the feedback resistors, their resistances are controlled by the AGC signal. The transimpedance gain is specified by the current mirror ratio (g_{m5}/g_{m4}) and the feedback resistance. M9 acts as a feedback transistor placed directly across the input and output terminals of the current mirror (M4, M5). Using this direct feedback topology, M9 reduces the input impedance seen at the photodiode, improving TIA speed by the cost of the lower output swing.

In [71] the traditional TIA is replaced by a digital-to-resistance converter (DRC) consisting of a logarithmic resistor network and a digital multi-bit $\Sigma\Delta$ modulator as shown in Fig. 16. A parallel resistor, R_{par} , improves sensitivity and dynamic range by extending the minimum value of I_{bias} that is supported. The combination of the DRC and R_{par} enable continuous operation of the Light-to-Digital Converter (LDC) over approximately a three-order-of-magnitude range of I_{bias} . The DRC is controlled through feedback, which consists of a digital accumulator and a nonuniform quantizer based on a ladder inverter quantizer/amplifier/filter (LIQAF) [71]. The variable resistance $R_{DRC}(t)$ is achieved with a set of PMOS switches connected to taps of a polysilicon resistor string ladder, with the PMOS switches being controlled by the output of a $\Sigma\Delta$ digital modulator that provides high-resolution resistance (32 steps from 64 K to 10 M Ω) control through dithering. In contrast to the previous AGC approaches, the LDC system offers a high digital integration that minimizes analog complexity.

B. DC Photocurrent Rejection Circuits

The PPG signal has a large DC component, which can reach 99% of the total signal value. A DC photocurrent

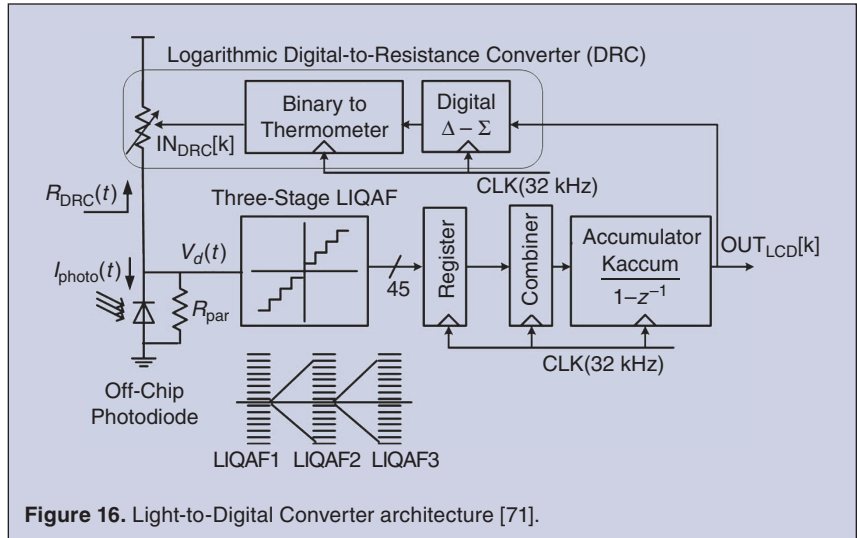


Figure 16. Light-to-Digital Converter architecture [71].

rejection is needed to prevent PPG receiver saturation. An optical preamplifier with ambient photocurrent rejection is shown in Fig. 17 [72]. The transimpedance amplifier converts the current from the photodiode into a differential voltage. Surrounding the amplifier is an outer feedback loop comprising an error amplifier and a transistor. The rejection circuit operates as follows: The DC component of the photocurrent produces an offset in the average levels of the differential outputs. This offset is integrated over time by the error amplifier. Transistor M_{ctl} acts as a variable current sink that, at steady-state, eliminates the average photocurrent from the signal path. The average photocurrent consists of the ambient photocurrent and the DC component of the signal. The transistor M_{10} in Fig. 15 [67] has the same function as M_{ctl} . For the pulsed PPG signal used to reduce sensor power consumption, the traditional method

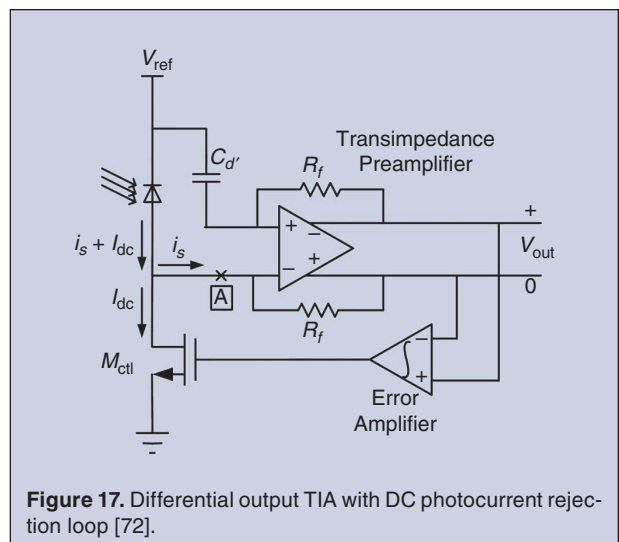


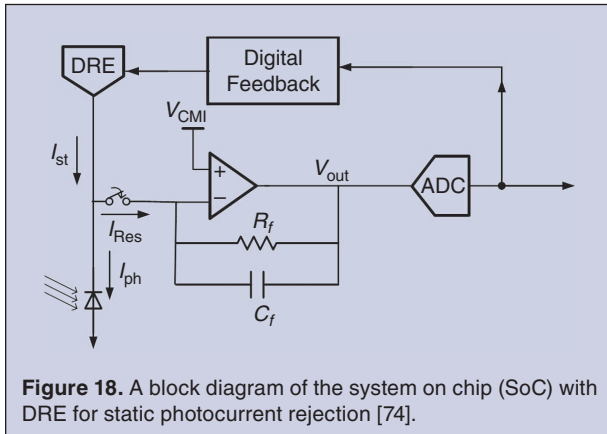
Figure 17. Differential output TIA with DC photocurrent rejection loop [72].

Using a differential signal path helps maximize the preamplifier immunity to noise from the power supply and substrate.

[68], [72] is not accurate. A DC photocurrent rejection TIA with sample-and-hold (S/H) in the feedback loop was introduced in [73]. The additional S/H block helps to detect the pulsed signal average accurately. Using a differential signal path helps maximize the preamplifier immunity to noise from the power supply and substrate. The photodiode, however, is connected to only one terminal. This creates an asymmetry at the input of the differential structure. As a result, an additional capacitor is required at the other input of the transimpedance amplifier in order to match the photodiode capacitance and to rebalance the differential circuit. Perfect matching ensures that noise injected at the bias voltage appears as a common-mode signal that is effectively rejected by the differential structure. In practice, however, some mismatch can be expected, and additional measures such

as adding an on-chip tuning capacitor may be required to improve matching.

A block diagram of the implemented PPG SoC system introduced in [74] is shown in Fig. 18. It consists of an integrate-and-hold amplifier, a dynamic range enhancement (DRE) current source, a 9-bit successive approximation (SAR) ADC, and a digital core to supply feedback and control signals to the analog blocks (not shown for simplicity), such as the sampling switch at the negative terminal of the op-amp and the reset switch across the CF [74]. The DRE circuit generates a current (I_{st}), which subtracts the static component of the photocurrent (I_{ph}), and the residue ($I_{ph} - I_{st}$) is integrated by the integrate-and-hold amplifier. Clean PPG data can be accessed from the SoC by using a standard SPI protocol, allowing for easy interfacing with low-power microcontrollers. However, this DRE technique is very complex and requires more area for the digital part compared to the analog solutions.

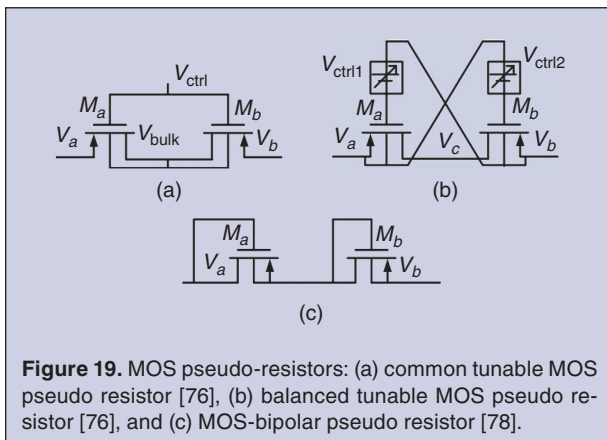


C. PPG Low Frequency Contents

The PPG signal has a low frequency of about 0.5 Hz and a high frequency up to 5 Hz. Therefore, the AGC and DC photocurrent cancellation circuits need to be designed with large time constants ($T = 1/RC$) to have less than a 0.5 Hz low cutoff frequency. In addition, the bandwidth of the PPG amplification stage should be around 5 Hz to reduce the out of band noise. This implies that large capacitances and large resistances must be implemented. The total area of the resistor and capacitor will be quite large. Some special techniques have been explored to solve the above problems.

1) MOS pseudo-resistor

A resistance of hundreds of mega ohms can be achieved when a MOSFET is biased in the subthreshold region. This type of MOS resistor is commonly used together with a capacitor in biomedical applications for low cutoff frequency [75]. Since tuning a resistor by trimming is inconvenient and costly, a tunable MOS pseudo-resistor was proposed [76], as shown in Fig. 19(a). The tolerable DC input voltage range was measured to be 0.5 V. Considering that this MOS resistor exhibits asymmetric and nonlinear resistance as the voltage varies across it, [77] proposed a balanced tunable MOS pseudo-resistor in their programmable biomedical sensor interface chip, as illustrated in Fig. 19(b). The two transistors can be turned on in an alternating manner



to achieve a symmetric incremental resistance curve. The resistance was claimed to be almost constant when the voltage across it changed. In addition, a MOS-bipolar device, which is a drain-gate shorted MOSFET, has also been proposed for implementing large resistances. Using a PMOSFET as an example, with negative voltage, it functions as a diode-connected PMOSFET; with positive voltage, the parasitic BJT is turned on and works as a diode-connected BJT. [78] employed this technique to design a circuit with a low cutoff frequency. Two MOS-bipolar devices connected in series were used to reduce the distortion of large output signals, as shown in Fig. 19(c). The cutoff frequency of the amplifier was 0.025 Hz, in which the equivalent resistance of the device was larger than $10^{13} \Omega$ [75]. However, standard MOS-bipolar pseudo resistors suffer from process, voltage, and temperature (PVT) variations in addition to possible light and EMI sensitivities, leading to variations in cutoff frequency.

2) Switched capacitors

Switched capacitors can be used to implement on-chip PVT-insensitive high resistance as shown in Fig. 20(a) [79]. In this topology, the switching frequency f_s and the capacitor in the middle determine the resistance precisely as $1/f_s C$. The switched-capacitor resistor in Fig. 20(b) mitigates manufacturability and interference issues by realizing a tenfold resistance increase by charge sharing in the switched-capacitor circuits [80]. Unfortunately, the switching produces additional noise along with the weak analog signal in the front end, decreasing the receiver sensitivity.

3) Miller capacitance multiplier

A Miller capacitance multiplier is used to implement a large effective capacitance value using a small value capacitor [81], [82]. A two-stage Miller capacitance multiplier for compensation is proposed to implement the high equivalent capacitance required for an ultra-low frequency biomedical amplifier [81]. In [83], using a capacitance multiplier, the TIA achieves a passband gain of 89 dBV, a DC current rejection ranging from 0.5 to 85 μA , and a 23 dB high pass cutoff frequency of 0.5 to 9.5 Hz. This corresponds to an improvement of six times the DC current rejection range compared with previous techniques [83]. The current through C_{m1} is converted to voltage by R_b , amplified and fed back to node Y, as shown in Fig. 21. The equivalent capacitance seen from node Y can be expressed as follows:

$$C_{eq} = \frac{g_{mcn1} r_{ox} g_{mcn2} R_b A_{v2}}{1 + s C_{m1} R_b} \cdot C_{m1}, \quad (9)$$

A disadvantage of such capacitance multiplication techniques is the limited linear range they possess, which is inversely proportional to the multiplication ratio.

4) α -block

To design a filter architecture with an ultra-low cutoff frequency, robust performance and compactness, active-RC implementations are preferred. The architecture of the first order current-steering low-pass filter (CS-LPF) is given in Fig. 22, and the α -block is enclosed within the

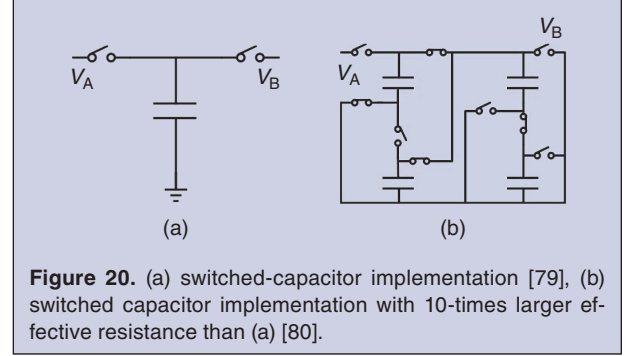


Figure 20. (a) switched-capacitor implementation [79], (b) switched capacitor implementation with 10-times larger effective resistance than (a) [80].

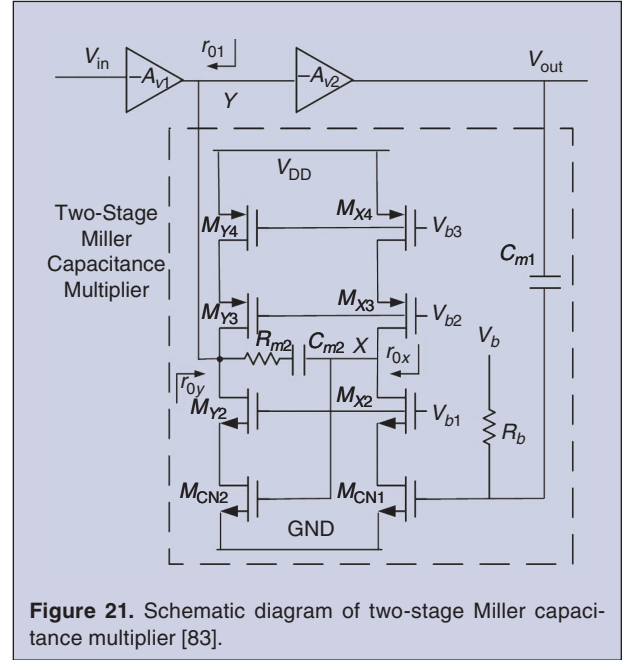


Figure 21. Schematic diagram of two-stage Miller capacitance multiplier [83].

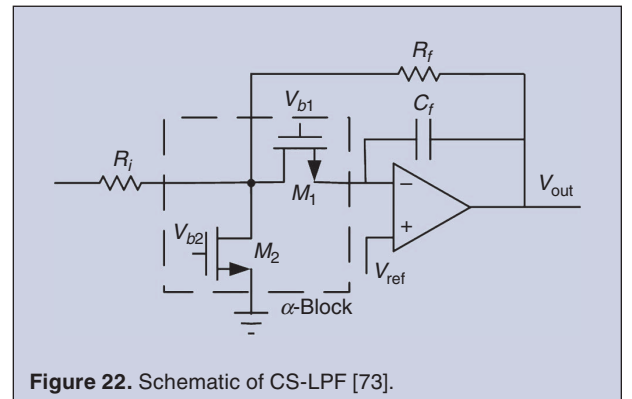


Figure 22. Schematic of CS-LPF [73].

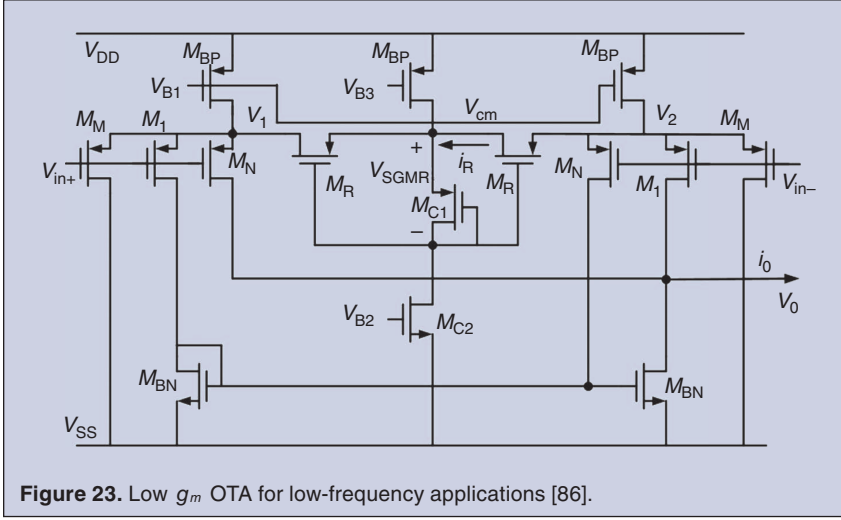


Figure 23. Low g_m OTA for low-frequency applications [86].

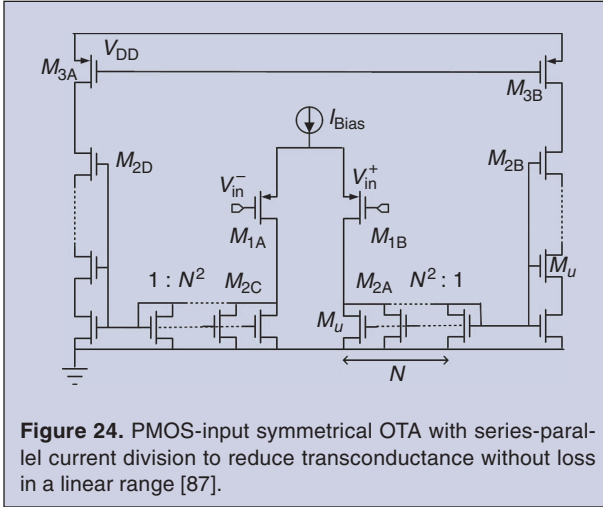


Figure 24. PMOS-input symmetrical OTA with series-parallel current division to reduce transconductance without loss in a linear range [87].

dotted line [73]. Here, the virtual ground at the negative input of the opamp has been extended to node X since the α -block is enclosed within a loop. For PPG sensing application, a single-ended version is sufficient. The new transfer function is given by the following:

$$A(s) = \frac{-R_F/R_i}{1 + \frac{SR_F C}{\alpha}}, \quad (10)$$

$$\alpha < 1$$

$$f = \frac{\alpha}{2\pi R_F C}, \quad (11)$$

With this technique, CS-LPF is implemented using a 38 pF capacitor to achieve a cutoff frequency from 62 to 1.8 Hz (and even lower). The effective capacitive reduction is from 500 to 12500 times [73].

The α -block consists of two transistors, with M_1 operating at the triode region and acting as a linear resistor

and M_2 in the saturation region. Both of them have identical aspect ratios but different gate voltages to steer current (shown in Fig. 22), where α is given by I_{d1}/I_{d2} . In [84], a low cutoff frequency control block is used, which consists of a transconductance block g_{m2} and a current-to-voltage block $1/g_{m1}$, where α is given by $g_{m2}/g_{m1} < 1$.

5) Small transconductance OTA

The LPF can be designed using a Gm-C technique. A reduced transconductance (g_m) OTA suitable for low-frequency applications is required. Several techniques have been proposed to implement small

g_m for designing these filters with low cutoff frequencies [75]. The current division technique has been proposed to reduce the small signal transconductances in voltage-to-current converters [85]. The currents of two MOSFETs are partially canceled at the output. The small signal transconductance of the OTA can be reduced to 10^{-8} A/V. The current division method is insufficient to implement transconductances at the nV/A level. To achieve an even smaller transconductance, the above methods can be combined [86], refer to Fig. 23. The transconductance of the OTA can be reduced to 10^{-9} A/V. The drain currents of M_1 and M_N are partially canceled at the output of the OTA, leading to very large transconductance reductions. It can be shown that the resulting small signal transconductance becomes the following:

$$G_m = \frac{i_0}{v_1 - v_2} = \frac{N-1}{M+N+1} g_{mMR} \quad (12)$$

In equation (12), M is defined as the ratio of transconductances of M_M and M_1 , while N is the ratio of transconductances of M_N and M_1 .

Another technique to reduce the OTA g_m is the series-parallel (SP) current division structure, which reduces the transconductance based on a SP current mirror. The transconductance of the OTA can be reduced to 10^{-12} A/V [87], indicated in Fig. 24. For the NMOS current mirrors in Fig. 24, N unit transistors are placed in series and in parallel to achieve an effective output transconductance $G_m = g_{m1}/N_2$, where g_{m1} is the gate transconductance of the transistors M_1 .

D. Modulation

Sources of additive noise to the PPG signal include systematic circuit offsets, flicker noise, external light sources interference, and 50 Hz power line noise. As PPG signals have a

With a modulated light source, the receiver can effectively suppress the interference from ambient and 1/f noise.

low bandwidth (5 Hz), higher frequency additive noise can be removed with low-pass filtering. However, static (DC) offsets and low frequency noise (also known as fixed pattern noise), which occupy the same spectrum as the PPG signal, require different rejection techniques. One solution is to modulate the light source to shift it to a higher spectrum range where there is less noise content, as shown in Fig. 25. With a modulated light source, the receiver can effectively suppress the interference from ambient and 1/f noise. Demodulation envelope detection is attractive due to the simplicity of its implementation and low power use, but it is prone to distortion by additive noise. The system illustrated in Fig. 25 uses decimation by 4 FIR filters for the demodulation. Down-sampling the signal by a factor M results in placing a copy of the signal back in the baseband, completing the demodulation [74]. For demodulation using multiplications inside the MCU [88], as shown in Fig. 26, the MCU spent much time in full-power processing mode, which led to an average high power consumption. To reduce power consumption in the MCU, the percentage of time that the processor core is in a sleep or power-down state should be increased. Processing time can be reduced by choosing an alternative MCU core that supports hardware multiplication [88].

Even though light modulation will save the MCU and LED transmitter power dissipation, the detection with the pre-amplifier requires a higher bandwidth and hence higher bias current than if modulation was not employed.

E. LED Driver

A key characteristic for the PPG LED driver circuit is its low noise; the optical receiver will amplify

any in-band transmitted noise reducing the SNR. Therefore, low-noise and low-ripple LED driver is required. For wearable/mobile devices, the battery lifetime is a critical issue. The battery lifetime can be extended by reducing the LED ON time; also, the LED current should be reduced as much as possible, as shown in Fig. 27. The sampling rate needs to be high enough not to miss the

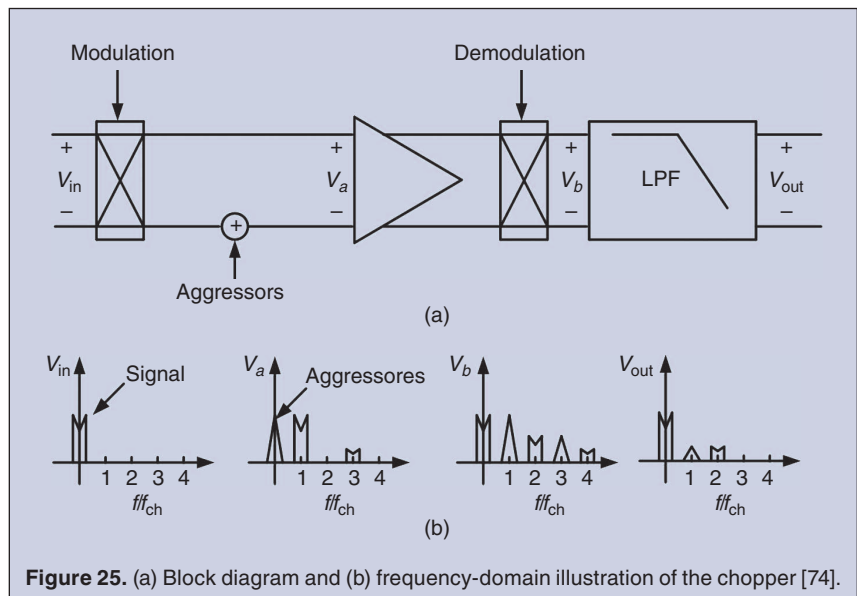


Figure 25. (a) Block diagram and (b) frequency-domain illustration of the chopper [74].

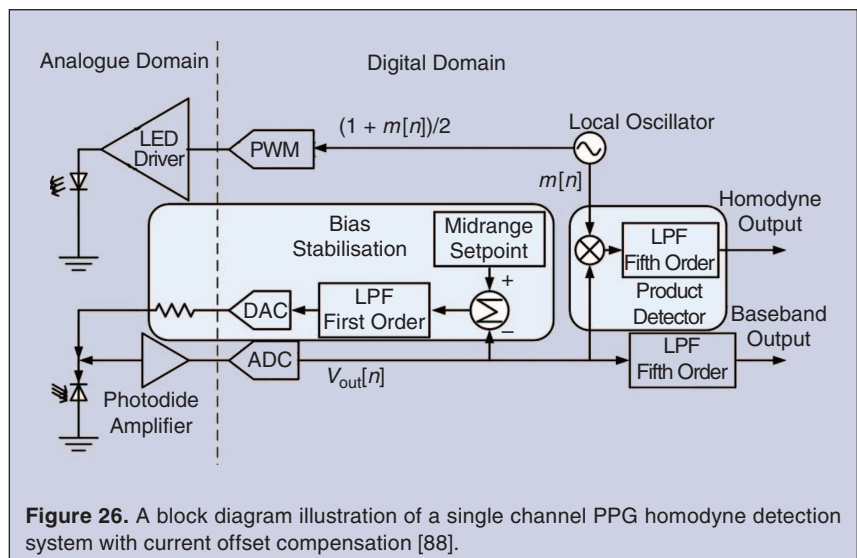
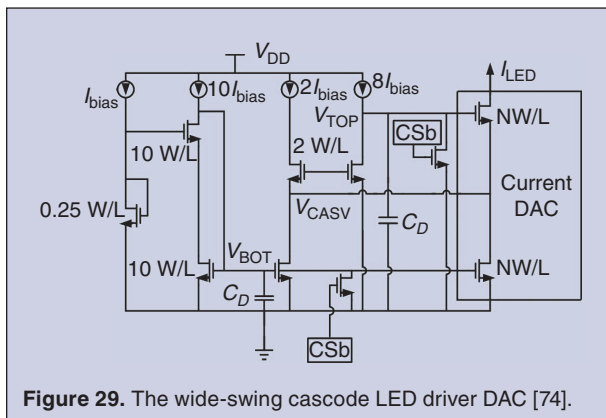
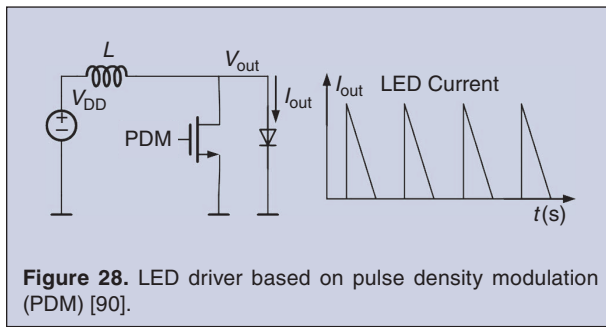
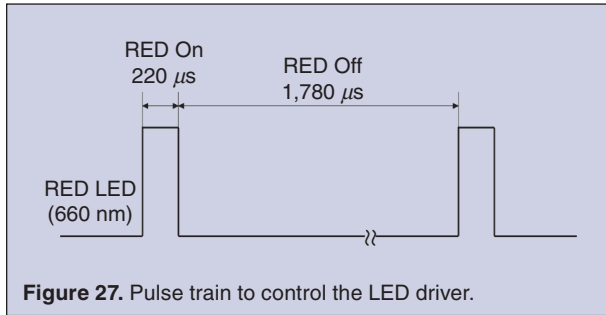


Figure 26. A block diagram illustration of a single channel PPG homodyne detection system with current offset compensation [88].

fast features of the PPG signal. Increasing the pulse repetition rate of the LED current results in better signal accuracy but at higher energy consumption. The LED current pulse repetition rate and pulse width are a trade-off between signal quality and the total consumed energy.



Typical systems sample with rates between 100 Hz and 1000 Hz with a pulse width ranging from 10 μ s up to 100 μ s. A study was introduced in [89] to select the optimum LED duty cycle to reduce the power consumption for a certain SNR.

A LED driver is proposed in [90] with 89% efficiency. A low supply voltage and low average current are desired for devices such as a PPG heart rate monitor, as shown in Fig. 28. The LED current and as a result the transmitted optical power is controlled by pulse density modulation (PDM). The integrated digital blocks for varying the LED optical power help to reduce the chip area. The LED driver implemented in 180 nm CMOS along with an external inductor has a controllable current range from 28 μ A up to 1.3 mA using a 0.5 to 0.6 V supply voltage.

The LED driver was designed as 8-bit, wide-swing cascode current mirrors, as shown in Fig. 29 [74]. This topology was chosen to maintain the high linearity of the output current across a large variation of the cascode drain voltage. Voltage variation occurs because the forward LED voltage can be as small as 1.1 V for an IR LED and as large as 1.8 V for a red LED. Additionally, for operation with a small coin-cell battery, the voltage can be as high as 3 V when the battery is fully charged and as low as 2.3 V when the battery discharges. The LSB of the LED driver DAC is 100 μ V with a maximum current of 25.6 mA. To reduce the voltage drop along the high current path, the DAC is toggled by switches connecting the gates of the cascode transistors to either V_{TOP} and or V_{BOT} to ground (not shown). A digitally controlled fast startup current reference for duty cycling the LED driver bias is also utilized to conserve the power. The duty cycle is 0.7%, which corresponds to a square LED current pulse that is 40 μ s wide every 60 ms.

F. PPG Integrated Sensor

A wearable cNIPB requires both low power and light weight/small area. The integration of the PPG receiver and the LED driver reduces the number of off-chip elements to obtain a compact portable system. Table IV

Table IV. Recently published integrated PPG sensors.					
	Process	Biasing	Noise	Rejection	Integrated LED Driver
[84]	0.35 μ m	2.5 V/58 μ A	40.8 nA	15 μ A, 165 pF on chip	No
[73]	0.35 μ m	2.5 V/240 μ A	3.53 nA	43.9 μ A, 2.2 nF Off chip	No
[71]	180 nm	0.5 V/8 μ A	4 nA	3.5 μ A logarithmic DRC	No
[74]	180 nm	2.3 V/425 μ W	2.5 nA	100 μ A, DRE current source	LSB 100 μ A maximum current of 25.6 mA
[70]	0.35 μ m	3.3 V/1.88 mW	445 μ V	No	1.3 mW, Feedback path to control the LED current

shows recently published PPG sensors. The PPG sensors in [71], [73], [74], [84] have a DC Photocurrent Rejection loop. The record of 100 μ A DC photocurrent rejection was achieved in [74] because of the (DRE) current source. Two recent PPG sensors [70], [74] have also integrated the LED driver on the same chip.

V. Commercial cNIBP Devices

There are many cNIBP devices based on PPG that are trying to find their way to the market. The ViSi Mobile system enables cNIBP of patients based on the PAT [91]. The SOMNOtouch NIBP simultaneously records the ECG and finger PPG, and the PTT is calculated. By means of a one-time calibration, the systolic and diastolic blood pressure can be calculated [92]. These commercial devices have an acceptable accuracy under small motion (rest condition) to reduce the motion artifact effect on the PPG signal. The processing is done in a separate PC unit to perform the BP analysis and calculations. The Simband smart watch from Samsung generates 8 PPG streams; based on the confidence indicator of the PPG channel, an algorithm may choose the best PPG. The ECG and xyz accelerometer signals are also available. The PAT is computed and can be used to calculate blood pressure. The Monitoring mode consumes a lot of power (400 mW peak power), as all the sensors and the processor are ON. This is not feasible in a system that need to work for 24 hours [93]. The LED ON time can be controlled to reduce the average power to 50 mW. The Project Zero Wrist Blood Pressure Monitor (BP6000) is a cuffless blood pressure smart watch developed by Omron [94]. The BP6000 results were shown at the CES 2016 event, and it will be available in the market shortly after Omron receives a certification from the FDA.

VI. Conclusion

Non-invasive, continuous time, cuffless BP devices have some major challenges to come to the market, with MAD levels required to be less than ± 5 mmHg. A calibration process is needed to reach that accuracy. More research is needed on MA cancellation algorithms to reach a true wearable, continuous time, cuffless BP device. The existing capabilities for the expected commercial PPG-based BP devices have acceptable performance under only very small motion effects. For beat-to-beat BP measurements, real-time processing is needed, which will require developing accurate BP algorithms with fewer operations and lower complexity. The most mature commercial BP devices deploy a wristwatch structure because it is comfortable and easier to wear, but the wrist PPG signal is weak. Therefore, more effort should be spent toward developing a high-sensitivity PPG receiver

to overcome the low wrist PPG signal. Light modulation can enhance the PPG receiver sensitivity at the expense of higher power consumption. Low power operation and small system area are necessities for a wearable device for cNIBP monitoring. Much power can be saved on the transmitting side by transmitting less optical power using a highly sensitive PPG receiver. The LED driver pulse repetition rate and pulse width should be optimized to have good signal quality and acceptable power consumption. To reduce the system area, the large capacitance/resistances required for the LPF and DC photocurrent cancellation can be integrated on chip by using several techniques such as a Gm-C LPF with nA/V transconductance OTA, capacitance multiplier, pseudo resistor, and α -block. The integration of the photodiode with the TIA and an LED driver using standard CMOS technology will reduce the system area and cost. Both smartwatches and smartphones are perfect candidate devices to have cNIBP measuring capability.

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